Visual Recognition using Deep Learning

Course Logistics

林彥宇 教授
Yen-Yu Lin, Professor
國立陽明交通大學 資訊工程學系
Computer Science, National Yang Ming Chiao Tung University

About Yen-Yu Lin

- Work experience
  - Professor, *CS, NYCU,* August 2019 ~ present
  - Associate research fellow, *CITI, Academia Sinica*, 2015 ~ 2019
  - Assistant research fellow, *CITI, Academia Sinica*, 2011 ~ 2015

- Research interests
  - Computer Vision (**CV**):

    Let computers see, recognize, and interpret the world like humans
  - Machine Learning (**ML**):

    Provide a statistical way to learn how human visual system works
  - Goal: *Design **ML** methods to facilitate **CV** applications*

Today's agenda

- Course logistics
- Course overview

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How to choose and take this course?

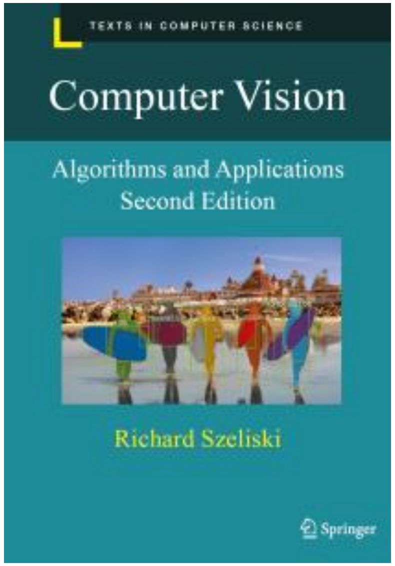
- Please use the online course management system
  - Max number: 100 students
- I do not plan to add additional students
  - Final project presentation arrangement
  - The size of the classroom
  - The loading of our TAs
  - If you have some strong reason why you must take this course, send me an email with the reason
- Be a guest student?
  - Yes. Send TAs an email with your student ID. We will add you to the student list on E3

Instructor and teaching assistants

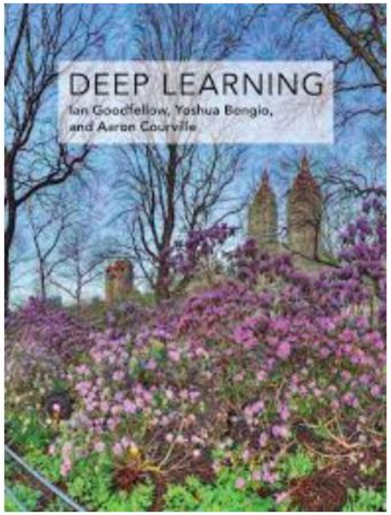
- Instructor: Yen-Yu Lin 林彥宇
  - Email: lin@cs.nycu.edu.tw
  - Office: EC706 (please email me first)
- Teaching assistants:
  - Tsung-Lin Tsai 蔡宗霖 Email: sean19990323123@gmail.com
  - Jian-Zhe Wang 王健哲 Email: jzwang.cs13@nycu.edu.tw
  - Yi-Jen Tsai 蔡宜蓁 Email: tsai.cs14@nycu.edu.tw
  - Nai-Yun Hsiao 蕭乃云 Email: alllllvin21292@gmail.com
- Office hour (email first)
  - 4:20 pm ~ 5:20 pm on Tuesdays at EC234-C or EC701

Optional textbooks

- Computer Vision: Algorithms and Applications, 2nd ed.
  - Richard Szeliski
  - Springer, 2022
  - Free copy online at https://szeliski.org/Book/



- Deep learning
  - I. Goodfellow, Y. Bengio, and A. Courville
  - MIT Press, 2016
  - Free online at https://www.deeplearningbook.org/



Grading policy

- Four homework assignments: 64% = 16% x 4
- Final project: 36%

- For each assignment
  - It is due at 11:59pm on the due day
  - Late policy: 20% off per late day
  - Evaluation criteria: model performance (70%), report (20%), and code readability (10%)

- For the final project
  - Team up for the final project
  - Find your partners by yourselves to yield a group of size 4

Grading policy

- Grade adjustment: I might further adjust the grade distribution to satisfy at least one of the following two conditions
  - Approximately 25% of students receive an A+
  - Approximately 50% of students receive an A+ or A

Pre-requisite

- Linear algebra and calculus
- Programming experience such as Python
  - Python: We strongly encourage students who are not familiar with Python to study the following tutorial
  - http://cs231n.github.io/python-numpy-tutorial/
- Deep learning programming skills
  - Pytorch: https://pytorch.org/tutorials/
  - Keras: https://elitedatascience.com/keras-tutorial-deep-learning-in-python
- It would be better to have a GPU card

Syllabus

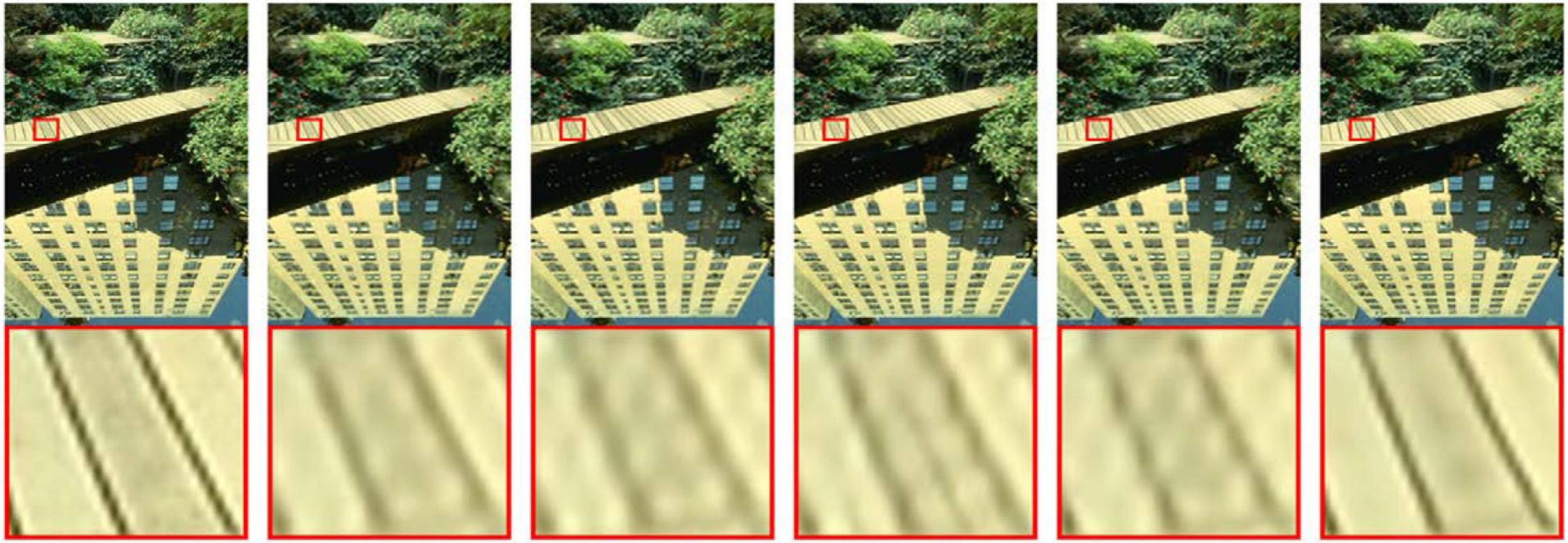
	1	2026-02-24(二)	Introduction
	2	2026-03-03(二)	Deep Neural Networks
HW1	3	2026-03-10(二)	Convolutional Neural Networks
	4	2026-03-17(二)	Transformers
	5	2026-03-24(二)	Object Detection I
HW2	6	2026-03-31(二)	Object Detection II
	7	2026-04-07(二)	Object Segmentation I
	8	2026-04-14(二)	Object Segmentation II
HW3	9	2026-04-21(二)	Denoising Diffusion Models
	10	2026-04-28(二)	Low-level Vision
Final Proj. Ann.	11	2026-05-05(二)	Mamba
HW4	12	2026-05-12(二)	3D Point Clouds, Neural Radiance Fields (NeRF), and 3D Gaussian Splatting (3DGS)
	13	2026-05-19(二)	3D Vision
	14	2026-05-26(二)	Guest Lecture (Date subject to change based on speakers' schedule)
	15	2026-06-02(二)	Final Project Presentation I
	16	2026-06-09(二)	Final Project Presentation II

Homework assignment 1: Car brand classification

Homework assignment 2: Digit detection

Homework assignment 3: Instance segmentation

Homework assignment 4: Super-resolution



Ground Truth	A+ [22]	RFL [18]	SelfEx [11]	SRCNN [5]	VDSR (Ours)
(PSNR, SSIM)	(22.92, 0.7379)	(22.90, 0.7332)	(23.00, 0.7439)	(23.15, 0.7487)	(23.50, 0.7777)

國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Thank You for Your Attention!

Yen-Yu Lin (林彥宇)

Email: lin@cs.nycu.edu.tw

URL: https://www.cs.nycu.edu.tw/members/detail/lin